

Problem 11.4

If $\delta_t = 0.01t$, $0 \leq t \leq 2$, find the equivalent annual effective rate of interest over the interval $0 \leq t \leq 2$.

Solution

The accumulation factor over the two-year interval is

$$\exp \left(\int_0^2 \delta_t dt \right).$$

First, evaluate the integral:

$$\begin{aligned} \int_0^2 \delta_t dt &= \int_0^2 0.01t dt \\ &= \left. \frac{0.01}{2} t^2 \right|_0^2 \\ &= 0.02. \end{aligned}$$

Let i be the equivalent annual effective rate. Over two years,

$$(1+i)^2 = e^{0.02}.$$

Taking the square root and solving for i gives

$$\begin{aligned} i &= e^{0.02/2} - 1 \\ &= e^{0.01} - 1 \\ &\approx 0.010050167. \end{aligned}$$

Hence,

$$\boxed{i \approx 1.0050\%}.$$